ACTIVERAG: Revealing the Treasures of Knowledge via Active Learning

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Abstract

Retrieval Augmented Generation (RAG) has introduced a new paradigm for Large Language Models (LLMs), aiding in the resolution of knowledge-intensive tasks. However, current RAG models position LLMs as passive knowledge receptors, thereby restricting their capacity for learning and comprehending external knowledge. In this paper, we present ACTIVERAG, an innovative RAG framework that shifts from passive knowledge acquisition to an active learning mechanism. This approach utilizes the Knowledge Construction mechanism to develop a deeper understanding of external knowledge by associating it with previously acquired or memorized knowledge. Subsequently, it designs the Cognitive Nexus mechanism to incorporate the outcomes from both chains of thought and knowledge construction, thereby calibrating the intrinsic cognition of LLMs. Our experimental results demonstrate that ACTIVERAG surpasses previous RAG models, achieving a 5% improvement on question-answering datasets. All data and codes are available at https://github.com/OpenMatch/ActiveRAG.

1 Introduction

Learners do not acquire knowledge and understanding by passively perceiving it, rather they construct new understandings and knowledge through experience and social discourse[1]*.*–Constructivism

Large Language Models (LLMs), *e.g.* GPT-4 (OpenAI, 2023) and LLaMA (Touvron et al., 2023), have shown strong emergent abilities and achieve convincing performance in various NLP tasks (Wei et al., 2022a; Zhao et al., 2023). Nonetheless,

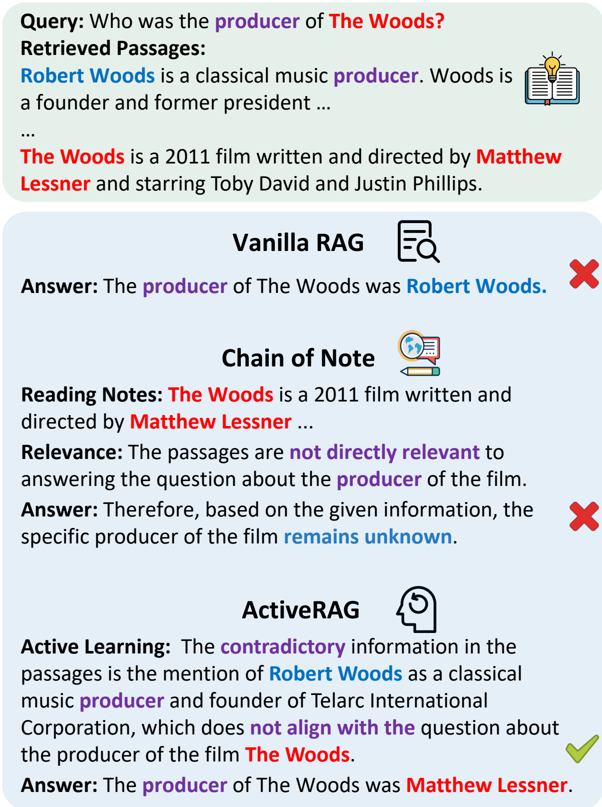

Figure 1: Illustrations of Different RAG Models. Vanilla RAG and Chain-of-Note (Yu et al., 2023b) employ passive learning processes, whereas ACTIVERAG involves active knowledge acquisition.

LLMs usually suffer from the hallucination problem (Ji et al., 2023; Bang et al., 2023) and outdated parametric memories (He et al., 2023), making them generate unreliable outputs (Si et al., 2022).

Recent research has proposed Retrieval-Augmented Generation (RAG) models (Guu et al., 2020; Lewis et al., 2020), showing their effectiveness in helping LLMs to update the learned knowledge, facilitating lots of tasks, especially these knowledge-intensive ones (Yu et al., 2023c). They retrieve knowledge from the external corpus and build a brute-force RAG

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[1]https://en.wikipedia.org/wiki/Constructivism_(philosophy_of_education)

architecture by directly feeding the passages to the LLMs. However, the effectiveness of retrieval-augmented models is highly influenced by the noise from retrieved knowledge (Chen et al., 2023). To conduct a more effective knowledge learning process, the work (Asai et al., 2023; Yu et al., 2023b) mainly focuses on mitigating the effect of noise from retrieved passages by self-reflection and self-refining. They try to filter out irrelevant passages (Asai et al., 2023) or conduct summarized noting (Yu et al., 2023b) to extract key point knowledge, which has shown effectiveness in assisting LLMs to generate more accurate answers for the given questions.

Despite the success of these self-refined RAG models, they still make the LLM a passive reception of external information, which violates the *Constructivism* (Steffe and Gale, 1995), a philosophy theory of education of humans. They neglect the nature of active learning, which builds associations between external knowledge and prior learned knowledge and constructs deeper knowledge understandings for LLMs. As depicted in Figure 1, both the vanilla RAG and Chain-of-Note models (Yu et al., 2023b) yield inaccurate answers, highlighting the limitations of passive knowledge acquisition. Specifically, the vanilla RAG model is misled by certain ambiguous entities, such as "The Woods" and "Robert Woods". Additionally, the Chain-of-Note model employs shallow relevance modeling, leading the LLM to produce unknown answers even if the summary note is correct.

Different from these RAG models, ACTIVERAG leverages active learning to augment knowledge comprehension by bridging the gap between prior knowledge and retrieved information, enabling the LLM to produce accurate answers. ACTIVERAG enhances the active learning capability of LLMs without necessitating further fine-tuning and employs a three-step pipeline to construct the RAG model, including Retrieval, *Knowledge Construction*, and *Cognition Nexus* stages. Specifically, ACTIVERAG initially retrieves passages from knowledge bases and subsequently *constructs knowledge comprehension* by associating the retrieved passages with pre-existing knowledge. Finally, the *Cognition Nexus* mechanism is utilized to connect external knowledge with the inherent cognition of LLMs. It first generates an initial chain-of-thought (CoT) (Wei et al., 2022b) for problem-solving and subsequently leverages the outcomes of knowledge construction to refine the CoT thereby producing more precise answers.

Our experiments utilize ChatGPT-3.5 to develop the ACTIVERAG model. Evaluation results demonstrate the effectiveness of ActiveRAG across various question answering datasets, yielding over a 5% improvement compared to baseline models. Furthermore, ActiveRAG exhibits robustness and maintains stable performance across different datasets and varying numbers of retrieved passages. Further analysis reveals that the knowledge construction outcomes can be generalized to different LLM architectures, such as LLaMA2-7B/13B, aiding them in leveraging external knowledge and achieving improvements exceeding 20%.

2 Related Work

Retrieval-Augmented Generation (RAG) models (Lewis et al., 2020; Guu et al., 2020; He et al., 2021; Xu et al., 2023; Borgeaud et al., 2022; Jiang et al., 2023) aim to retrieve external knowledge and enhance language models to overcome the limitations of purely parametric models. They show strong effectiveness in various NLP tasks (Ram et al., 2023a), such as question answering (Izacard et al., 2023), dialog understanding (Shuster et al., 2022), code generation (Zhou et al., 2022) and so on. RAG models typically utilize dense retrievers (Karpukhin et al., 2020; Xiong et al., 2020) as their knowledge-seeking modules, subsequently prompting language models to generate results based on the retrieved knowledge.

Earlier RAG models (Guu et al., 2020; Izacard et al., 2023) mainly focus on optimizing models to leverage external knowledge and better serve the knowledge-intensive tasks (Petroni et al., 2021). The work aims to develop the architecture of the generator to fully use external knowledge (Izacard and Grave, 2021), train a more accurate retriever using the feedback from generation models (Izacard and Grave, 2020), or jointly train the retriever and generator (Singh et al., 2021). With the development of Large Language Models (LLMs), researchers pay more attention to generalizing the advance of RAG to LLMs. They leverage external knowledge to reduce the generation perplexity of LLMs in general NLP tasks (Shi et al., 2023; Ram et al., 2023a) and alleviate the hallucination problem of LLMs by updating the out-of-date and long-tail knowledge (Kandpal et al., 2023; Shuster et al., 2021). Nevertheless, the noise of retrieved contexts usually challenges the effectiveness of RAG models (Yu et al., 2022; Chen et al., 2023).

In order to mitigate the impact of noise from retrieved contexts, prior research endeavors to eliminate irrelevant contexts (Xu et al., 2023; Asai et al., 2023), enhancing the robustness of the RAG model. Specifically, this work uses natural language inference models to select related sentences (Yoran et al., 2023b), a summarization model to pick up necessary semantics (Xu et al., 2023) or the conditional cross-mutual information to estimate the informativeness of context (Wang et al., 2023). Besides, Self-RAG (Asai et al., 2023) uses LLMs to filter out irrelevant passages via self-reflection.

Instead of finetuning LLMs to refine the retrieved results (Asai et al., 2023), recent work provides some promising ways to improve the generation accuracy of RAG models. They adaptively retrieve passages (Jiang et al., 2023) or build the noting mechanism to summarize the key point knowledge from retrieved passages (Yu et al., 2023b). Despite their success, LLMs continue to passively acquire additional knowledge and often overlook the *Constructivism* theory, which associates external knowledge with prior knowledge.

3 Methodology

In this section, we introduce our ACTIVERAG method, which guides LLMs to actively read, understand and use the external knowledge for generation. Generally, we first introduce the Retrieval-Augmented Generation (RAG) models (Sec. 3.1) and then describe our ACTIVERAG method that prompts LLMs to actively learn knowledge from retrieved passages (Sec. 3.2).

3.1 Preliminary of Retrieval-Augmented Generation (RAG) Models

Given a query q, existing Retrieval-Augmented Generation (RAG) models (Guu et al., 2020; Izacard et al., 2023) utilize retrieval models to search passages $D = \{d_1, ..., d_n\}$ and enhance the generation ability of language models by grounding these retrieved passages.

Vanilla RAG. Earlier RAG models usually employ the *retrieval-generation* architecture, which directly feeds the retrieval passages $D = \{d_1, ..., d_n\}$ to language models to generate the answers for the given query q. Nevertheless, these retrieved passages usually contain noise and the brute-force RAG approach tends to constrain the benefits of RAG modeling (Chen et al., 2023; Xu et al., 2023; Wang et al., 2023). It even sparks discussions on augmenting LLMs using the retrieval results or model-generated outputs (Yu et al., 2022).

RAG with Self-Refining. To avoid the effect of the noise from retrieved passages, recent work (Asai et al., 2023; Yu et al., 2023b) employs the *retrieval-refining-generation* architecture to empower the capability of RAG using LLMs.

Self-RAG (Asai et al., 2023) and Chain-of-Note (Yu et al., 2023b) are two representative methods to summarize knowledge from retrieved passages, depending on the self-reflection abilities of LLMs. Specifically, Self-RAG focuses more on finetuning LLMs to adaptively retrieve passages on-demand and controlling the information flows from retrievers to generators. Instead of finetuning, Chain-of-Note designs the instructions to self-refine the retrieved passages and forms the query-focused summarization, which can be applied to the black-box LLMs, such as GPT-3.5. Nevertheless, they neglect the nature of active learning–learner actively construct their understanding and knowledge by integrating new information with prior knowledge (Steffe and Gale, 1995).

3.2 ACTIVERAG: RAG with Active Knowledge Learning

Inspired by the constructivism learning theory (Steffe and Gale, 1995), we build a new RAG architecture, ACTIVERAG, to teach LLMs to actively acquire knowledge by themselves.

As shown in Figure 2, ACTIVERAG designs a three-step pipeline, including retrieval, knowledge construction, and cognitive nexus. Different from the self-refined RAG models (Asai et al., 2023; Yu et al., 2023b), ACTIVERAG focuses more on conducting active knowledge learning by bridging the gap between retrieved passages and priorly learned knowledge of LLMs without any finetuning.

Specifically, ACTIVERAG conducts different *Knowledge Construction* methods to form the knowledge understanding results from retrieved passages and then utilizes the *Cognitive Nexus* mechanism to integrate the acquired knowledge understandings with the raw chain-of-thought to assist LLMs for generation. All prompt templates used in ACTIVERAG are shown in Appendix A.2.

Knowledge Construction. To form the knowledge understanding from the retrieved passage, ACTIVERAG regards LLM as a cognitive structure of receiving, understanding, and transforming external knowledge from passages (Atkinson and

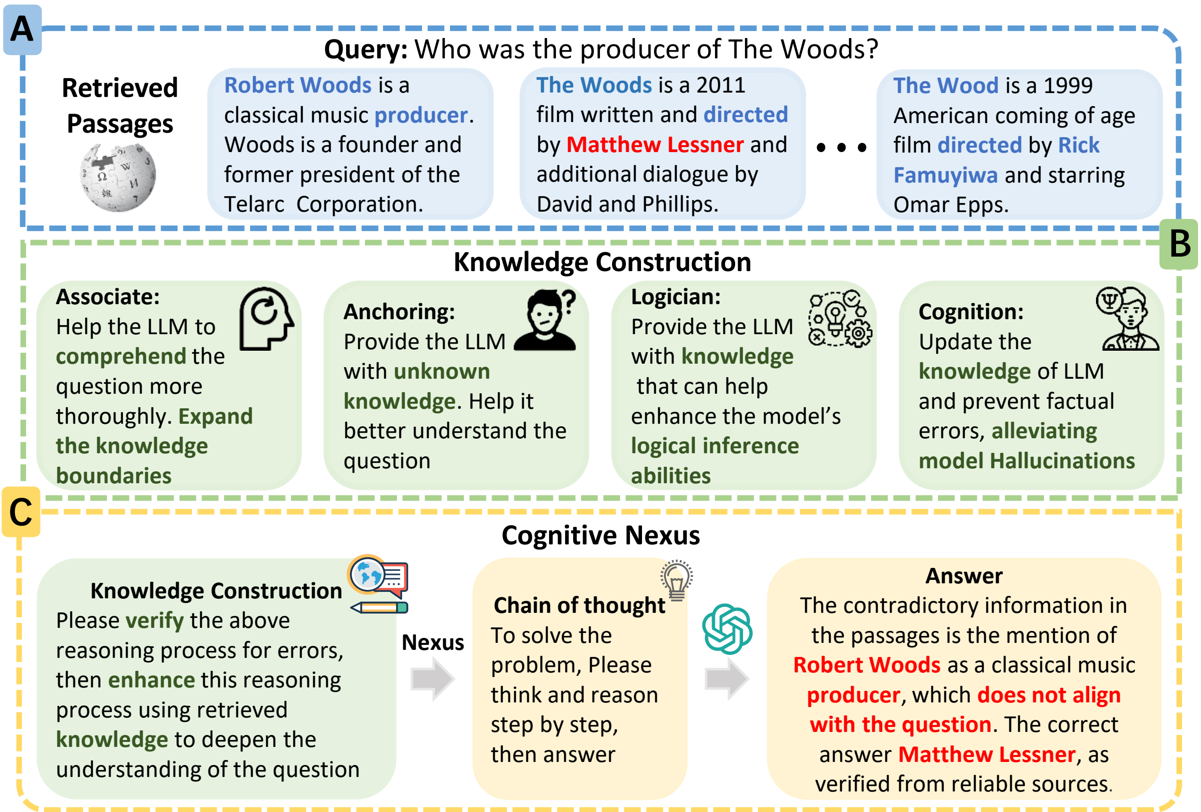

Figure 2: The Overview of Our ACTIVERAG Model.

Shiffrin, 1968). To mimic the learning behaviors of humans, we construct four distinct agents dedicated to knowledge construction and assess the proficiency of LLMs from various perspectives of knowledge acquisition. The perspectives encompass anchoring, logical reasoning, cognition, and association. We show the detailed descriptions of these knowledge construction agents in Sec. 3.3.

Cognitive Nexus. Following the constructivism learning theory (Steffe and Gale, 1995), we employ the cognitive nexus mechanism to assist LLMs in utilizing external knowledge for augmentation. The cognitive nexus mechanism facilitates the fusion of the constructed knowledge understanding with the intrinsic cognitive processes of LLMs. Specifically, we prompt LLMs to generate an initial chain-of-thought (Wei et al., 2022b) for the given query q, representing the cognitive process of LLMs. Subsequently, we integrate the outcomes of knowledge construction into the chain-of-thought to encourage LLMs to reflect on and rectify any factual errors it may contain. In contrast to RAG models, ACTIVERAG emphasizes the importance of incorporating the process of knowledge construction into self-aware cognitive reasoning, rather than solely focusing on refining passages.

3.3 Knowledge Construction from Different Learning Views

The Constructivism learning theory posits that learning is an active process involving the construction of internal mental representations that encompass both structural knowledge and extensive non-structural experiential backgrounds (Piaget, 1970). In order to facilitate active learning using memorized knowledge, we prompt LLMs as four agents to acquire knowledge from retrieved passages.

Semantic Association (Associate). The Semantic Association encourages the model to integrate familiar information and foster a deeper understanding through the consolidation of foundational and advanced pieces of information. Profound knowledge helps to expand the model's cognitive boundaries, enabling a more comprehensive understanding of the query and knowledge.

Epistemic Anchoring (Anchoring). This facet pertains to unfamiliar knowledge and focuses on representing information previously unknown to the model. The epistemic anchoring serves to establish a foundational understanding and incorporate new concepts relevant to the given question, allowing LLMs to actively analyze and construct knowledge from unfamiliar content.

Dataset	#Query	Sampling Rate
PopQA	14,267	3.5%
TriviaQA	8,837	5.7%
NQ	8,757	5.7%
WebQ	2,032	24.6%

Table 1: Data Statistics. Each dataset consists of randomly sampled 500 questions from the raw dataset.

Logical Reasoning (Logician). Constructivism learning theory suggests that learning is not merely about absorbing information but involves the active construction of knowledge through reasoning. Logical Reasoning leverages structured information to draw logical conclusions and refine the model's problem-solving capabilities.

Cognitive Alignment (Cognition). Constructivism recognizes that learners may need to adjust existing mental structures to incorporate new knowledge. The Cognitive Alignment deals with knowledge that may contradict the model's pre-existing understanding, preventing factual errors and mitigating the hallucination of LLMs.

4 Experimental Methodology

In this section, we describe the datasets, baselines, and implementation details in our experiments.

Dataset. As shown in Table 1, we use four datasets of open-domain question answering to evaluate the performance of ACTIVERAG, including Natural Questions (NQ) (Kwiatkowski et al., 2019), PopQA (Mallen et al., 2023), TriviaQA (Joshi et al., 2017) and WebQ (Berant et al., 2013). We randomly sample a subset of 500 questions from each QA dataset in our experiments, due to the inference cost of ChatGPT, in line with prior work (Trivedi et al., 2023; Yoran et al., 2023a).

Evaluation Metrics. We use accuracy (Acc) to evaluate the performance of different models in open domain QA, which is similar to previous work (Yu et al., 2023c). We first convert both the outputs of the LLM and golden answers to lowercase and subsequently perform string matching (StringEM) between each golden answer and the model prediction to calculate the accuracy.

Baselines. We compare ACTIVERAG with several baseline models, including prompt learning and RAG models. All prompt templates used by baselines are shown in Appendix A.3.

Prompt Learning. We first feed the QA instruction to ChatGPT to conduct the vanilla answer generation model. Then we follow previous work (Wei et al., 2022b) and implement the Chain-of-Thought model, which generates the question rationale results to produce the final results. Besides, we use the Guideline model as a baseline, which generates the problem-solving steps and guides LLMs to generate the answer.

RAG Modeling. For the RAG based baselines, we implement four baseline models, including vanilla RAG, Chain-of-Note, Self-Rerank and Self-Refine. The vanilla RAG model directly feeds the top-ranked passages to the LLM. Then we implement the Chain-of-Note (Yu et al., 2023b) model, which refines and summarizes the retrieved passages for generation. Inspired by Self-RAG (Asai et al., 2023), we also conduct the Self-Rerank and Self-Refine models, which filter out unrelated contents without finetuning LLMs. Different from Self-Rerank, Self-Refine further summarizes the reserved passages to assist the answer generation.

Implementation Details. We access GPT models via the OpenAI API to evaluate the performance of the ACTIVERAG. Specifically, we employ `gpt-3.5-turbo-1106` as the foundation model for inference and set the temperature at 0.2 during the generation. In our experiments, we use T5-ANCE (Xiong et al., 2020; Ge et al., 2023) to retrieve top-k relevant documents from KILT-Wikipedia (Petroni et al., 2021) for each question. The retrieval model is implemented by OpenMatch toolkit (Liu et al., 2021; Yu et al., 2023a).

5 Evaluation Result

In this section, we show the overall performance of ACTIVERAG. Subsequently, we delve into the analyses of the characteristics of various knowledge construction mechanisms and the generalization capability of knowledge construction outcomes. Finally, case studies are shown.

5.1 Overall Performance

The overall performance of ACTIVERAG on different open-domain QA datasets is shown in Table 2.

We compare ACTIVERAG with several baseline models, including LLMs w/o RAG, the vanilla RAG model and self-refined RAG models. Compared with the ChatGPT-3.5 model, the vanilla RAG model shows distinct performance on different QA datasets. Upon passage input into ChatGPT-3.5, a decrease in performance is observed on the WebQ dataset, while exhibiting nearly equivalent performance on the TrivialQA and NQ datasets, which has been observed in previous work (Yu

Method	Model	Top-5				Top-10			
		NQ	TriviaQA	PopQA	WebQ	NQ	TriviaQA	PopQA	WebQ
w/o RAG	ChatGPT-3.5	51.6	88.8	46.6	52.2	51.6	88.8	46.6	52.2
	CoT	53.6	89.4	45.0	56.6	53.6	89.4	45.0	56.6
	Guideline	54.4	88.2	45.0	46.2	54.4	88.2	45.0	46.2
Vanilla RAG	ChatGPT-3.5	48.6	82.8	54.4	36.4	53.2	86.6	57.6	42.4
Self-Refined RAG	Self-Rerank	47.6	83.6	53.6	35.6	48.6	83.0	56.2	37.6
	Self-Refine	46.6	80.6	53.2	34.8	48.6	79.6	56.4	36.0
	Chain-of-Note	56.4	86.2	64.6	51.4	61.8	88.0	69.0	54.0
ActiveRAG	w. Associate	**62.2**	**93.0**	**68.4**	**62.4**	**63.8**	**93.0**	**70.8**	**61.4**

Table 2: Overall Performance. In our experiments, we utilize the top 5 and top 10 ranked passages to evaluate the model's performance. All models are implemented using ChatGPT-3.5. We present the performance results of ActiveRAG, which is implemented using the Associate-based knowledge construction method.

Method	Setting	Top-5				Top-10			
		NQ	TriviaQA	PopQA	WebQ	NQ	TriviaQA	PopQA	WebQ
RAG	Vanilla	48.6	82.8	54.4	36.4	53.2	86.6	57.6	42.4
CoT	Vanilla	53.6	89.4	45.0	56.6	53.6	89.4	45.0	56.6
	w. Passage	61.2	91.0	61.0	59.4	60.4	91.2	62.2	59.4
	w. Note	61.2	91.8	56.2	60.2	61.8	91.0	58.0	61.0
ActiveRAG	w. Anchoring	60.2	92.4	63.6	59.8	61.6	92.2	64.4	59.0
	w. Logician	60.0	91.0	67.0	56.0	63.0	91.2	68.6	58.2
	w. Cognition	60.6	91.8	67.2	59.2	60.6	92.2	68.4	59.6
	w. Associate	**62.2**	**93.0**	**68.4**	**62.4**	**63.8**	**93.0**	**70.8**	**61.4**
	Oracle Rerank	69.4	95.2	75.0	66.6	70.0	95.2	79.0	67.6

Table 3: Ablation Studies. ActiveRAG designs different knowledge construction methods and we present the oracle reranking results of these knowledge construction methods to demonstrate the upper bound performance by integrating all these approaches. The **best** and second-best evaluation results are highlighted.

et al., 2022). After refining the passage to enhance LLMs, the performance of the RAG model shows some improvements. However, this refinement results in a decrease in the performance of ChatGPT-3.5 on both the TrivialQA and WebQ datasets. The main reason may lie in that passive knowledge acquisition fosters only superficial understanding, potentially misleading the LLM when reading passages with such a shallow understanding.

Compared to all baseline models, ActiveRAG surpasses them all with over 5% improvements, particularly excelling on the WebQ dataset, demonstrating its effectiveness. Notably, ActiveRAG consistently achieves significant improvements over the ChatGPT-3.5 model across all datasets, indicating the capacity of our method to guide LLMs in uncovering valuable knowledge from retrieved passages through active learning.

Additionally, we conduct experiments to present the RAG results using 5 and 10 top-ranked passages. Overall, all baseline RAG models consistently exhibit improvements when provided with more retrieved passages for generation. This indicates that additional related passages can serve as supplementary cues, aiding LLMs in refining their knowledge within these self-refined RAG models. On the contrary, ActiveRAG shows nearly identical performance when provided with either the top 5 or top 10 passages. This indicates that the information extracted from the top 5 passages is adequate to prompt LLMs to generate the correct answer. ActiveRAG proves to be effective in uncovering necessary knowledge from these retrieved passages with the help of knowledge construction.

5.2 Ablation Study

In our experiment, we conduct ablation studies to demonstrate the effectiveness of the cognitive nexus method by integrating different knowledge construction methods with the chain-of-thought.

As shown in Table 3, we process the retrieved passages with different methods and subsequently construct a cognitive nexus to aid LLMs. This involves several baseline models, including utilizing the raw retrieved passages (CoT w. Passage), refining the passages as notes (CoT w. Note), and employing the knowledge construction approaches of ActiveRAG. We further show the upper bound performance of these Knowledge construction methods through oracle reranking.

Our experimental results demonstrate that CoT w. Passage improves the performance of the vanilla RAG model, demonstrating the importance of establishing a cognitive nexus between external

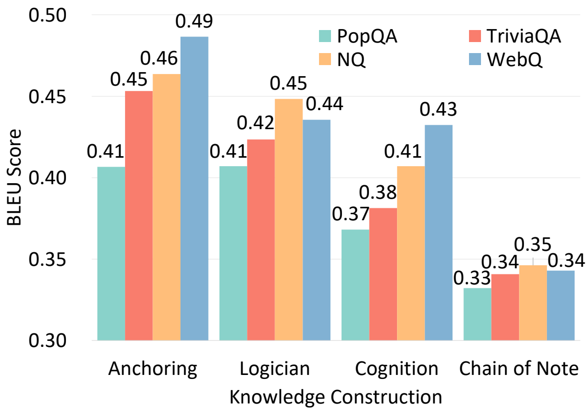

Figure 3: The Text Similarity between Associate and Other Knowledge Construction Method. In our experiments, we use the BLEU-2 score for evaluation.

Methods	NQ	TriviaQA	PopQA	WebQ
LLaMA2-7B				
Vanilla RAG	26.4	41.8	30.8	24.8
Chain-of-Note	22.0	45.2	38.1	21.0
CoT	32.2	66.0	25.2	41.8
LLM w/o RAG	26.8	53.2	21.6	40.2
w. Associate	50.6	87.4	55.8	49.6
w. Anchoring	52.4	**88.6**	51.6	**54.0**
w. Logician	51.2	<u>87.6</u>	**59.6**	50.2
w. Cognition	**54.4**	87.2	<u>58.4</u>	50.6
PPL-Rerank	<u>53.4</u>	<u>87.6</u>	56.2	<u>50.8</u>
LLaMA2-13B				
Vanilla RAG	29.0	39.0	22.8	25.2
Chain-of-Note	22.0	46.0	30.6	22.8
CoT	34.0	73.2	31.6	48.8
LLM w/o RAG	30.6	57.2	32.2	44.6
w. Associate	53.2	<u>89.6</u>	57.4	52.6
w. Anchoring	51.6	88.2	50.6	**53.8**
w. Logician	**57.8**	**90.3**	**65.4**	53.0
w. Cognition	51.2	88.6	52.4	51.6
PPL-Rerank	<u>55.2</u>	89.4	<u>61.8</u>	<u>53.2</u>

Table 4: Generalization Effectiveness of the Outcomes of Chain-of-Note and ACTIVERAG. The 5 top-ranked passages are used for augmentation. The **best** and <u>second-best</u> evaluation results are highlighted.

knowledge and the intrinsic cognition of LLMs. Additionally, CoT w. Passage significantly improves the quality of the generated outputs for the LLM that employs the CoT method. Moreover, it offers insights into mitigating the hallucination problem (Shuster et al., 2021; Lin et al., 2022) in LLMs by employing a cognitive nexus that refines the raw chain-of-thought with external knowledge.

Compared with raw/refined knowledge representation methods, ACTIVERAG demonstrates the best performance across all datasets, thus highlighting the efficacy of our knowledge construction methods. Notably, among various knowledge construction methods, Associate outperforms others, demonstrating the necessity of leveraging external knowledge to deepen the understanding of LLMs. Furthermore, we employ the oracle rerank method to illustrate the potential effectiveness of integrating different knowledge construction methods. The evaluation results show that there is still room to improve the performance of ACTIVERAG. It indicates that leveraging diverse knowledge construction methods potentially aids LLMs in answering different kinds of questions.

5.3 Characteristics of Different Knowledge Construction Mechanisms

As shown in Figure 3, we conduct experiments to show the characteristics of the Associate based knowledge construction method.

The evaluation results reveal that the Associate method exhibits less similarity to the chain-of-note approach, indicating a divergence in knowledge construction from simple summarization. The Associate method yields results more akin to Anchoring, suggesting that both mechanisms primarily extract knowledge from retrieved passages. In contrast, the results of the Associate method diverge more significantly from those of Cognition. It is evident that such counterfactual cognition typically leads to disparate cognitive outcomes, rather than focusing on query-relevant content. These observations collectively indicate that LLMs possess the capability to construct knowledge and understanding from diverse perspectives.

5.4 Generalization Ability of Knowledge Construction Outcomes

This experiment shows the quality of the knowledge construction outcomes using different LLMs. As shown in Table 4, we leverage the outcomes of ACTIVERAG to prompt weaker LLMs, specifically LLaMA2-7B/13B, for answer generation.

Compared with the LLM w/o RAG, both vanilla RAG and Chain-of-Note consistently exhibit decreased performance. This trend suggests that retrieved passages often become the noise that adversely impacts model performance, even with refined notes. The primary factor behind this phenomenon is that the LLaMA2-7B/13B model is misled by these retrieved passages, due to its limited capacity to analyze and locate knowledge. Conversely, the knowledge construction outcomes yield significant improvements of over 20% for LLaMA2-7B/13B across all QA datasets, demonstrating their effectiveness. The knowledge con-

Question: Which sports event was first held at Worcester, Massachusetts in 1927? **Answer:** The first Ryder Cup	
Chain-of-Thought: 1. Think about sports events that could have been held in Worcester, Massachusetts in 1927. 2. Consider the popularity of different sports at that time. 3. Try to recall any historical events or records related to sports in Worcester in 1927. **Answer:** The first American Motorcycle Association (AMA) dirt track race.	
Step 1: Passage Retrieval	
Passage 1: Worcester now hosts the Massachusetts Pirates, an indoor football team in the National Arena League ... The city previously was home to the New England Surge of the defunct Continental Indoor Football League. **Passage 2:** Worcester Country Club is a private country club. The course hosted the first Ryder Cup in 1927. One of three golf courses in the United States to host the men's and women's U.S. Hazeltine made it an elite group of three upon its host of the 2016 Ryder Cup. Worcester also hosted the first-ever U.S. Open qualifying round in 1924. **Passage 3:** Worcester, Massachusetts is home to minor league sports teams ... Many historic and local sporting events have occurred in Worcester such as the first official Ryder Cup golf tournament at Worcester Country Club in 1927. **Passage 4:** The first public basketball game was played in Springfield, Massachusetts, on March 11, 1892.	
Step 2: Knowledge Construction using Associate Agent	
Foundational Knowledge: Worcester, Massachusetts has a history of hosting various sports events, including golf tournaments, basketball games, and minor league sports teams. The Worcester Country Club hosted the first Ryder Cup in 1927, making it a significant location for the event ... *Advanced Information:* Details about specific sports events and teams that have been hosted or based in Worcester, such as the Worcester Busters, Worcester Boosters, and Worcester Panthers baseball teams ... *Analyzing the Role of Information:* Worcester, Massachusetts has a rich sports history and has been a significant location for various sports events and teams. The hosting of the first Ryder Cup in 1927 at the Worcester Country Club highlights the city's importance in the world of golf. Additionally, the presence of minor league sports teams and professional franchises demonstrates the city's active participation in a wide range of sports. This information deepens the model's understanding of the question by providing context about the sports culture in Worcester and expanding its knowledge boundaries about the specific events and teams associated with the city.	
Step 3: Cognitive Nexus	
1. Identify popular sports events in 1927, considering Worcester, Massachusetts as the location. 2. Worcester, Massachusetts has a rich history in sports, particularly in baseball and golf. The Worcester Country Club hosted significant events such as the first Ryder Cup in 1927 and the 1925 U.S. Open. 3. The Worcester Country Club also hosted the 1960 U.S. Women's Open and was the site of the first-ever U.S. Open qualifying round in 1924, further solidifying its importance in the sports world. 4. Considering the historical significance of the Worcester Country Club in hosting major golf events, it is likely that the sports event first held at Worcester, Massachusetts in 1927 was the first Ryder Cup.	

Table 5: Case Study. We present one case to show the pipeline of ACTIVERAG, which includes passage retrieval, knowledge construction and cognitive nexus steps. The answers and relevant contents are highlighted.

struction method diverges from the Chain-of-Note approach, instead establishing the knowledge understanding by connecting retrieved passages with previously acquired knowledge. The method embodies the teacher's (ChatGPT-3.5) thinking process and serves as a kind of cognitive roadmap to guide the student (LLaMA2-7B/13B) to learn knowledge from retrieved passages. Besides, we enhance the outcomes of various knowledge construction mechanisms by assessing their quality through the calculation of query-conditioned perplexity (PPL). Our evaluation results demonstrate that the LLM has the capability to select a more suitable knowledge construction method for obtaining precise answers.

5.5 Case Studies

Finally, we randomly sample one case in Table 5 to show the effectiveness of ACTIVERAG. More cases are shown in Appendix A.4.

Our knowledge construction mechanism shows its effectiveness in forming the knowledge understanding results with learned knowledge, such as foundational and advanced information. The extracted information offers essential clues for answering questions. Moreover, the analysis of knowledge construction provides brief summaries and illustrates the application of this external knowledge. By incorporating the knowledge construction outcomes, our cognitive nexus module calibrates the raw chain-of-thought results and generates the correct answer. Specifically, the related knowledge from passages, such as "sports events in 1927", "golf", and "1925 U.S. Open", makes the thought more knowledgeable and detailed. This demonstrates the effectiveness of ACTIVERAG in supporting LLMs to combine both external knowledge and intrinsic cognition for answer generation.

6 Conclusion

This paper proposes ACTIVERAG, which constructs the retrieval-augmentation architecture incorporating active knowledge learning. Inspired by the Constructivism theory, ACTIVERAG builds the knowledge construction and cognitive nexus mechanisms to facilitate the integration of retrieved knowledge with the intrinsic cognition of LLMs.

Limitation

While ACTIVERAG proves to be an effective approach in harnessing external knowledge to assist LLMs in generation tasks without finetuning, it necessitates calling the ChatGPT API thrice: once to construct the initial chain-of-thought, again to process the knowledge construction results, and finally to execute the cognitive nexus for generating the ultimate answer. It may bring additional time latency and API calling costs to deal with the questions. Moreover, inputs to LLMs tend to be lengthy due to the inclusion of extensive retrieved passages and knowledge construction results.

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A Appendix

A.1 License

For all datasets utilized in our experiments, Triviaqa and WebQ are governed by the Apache License 2.0, PopQA is subject to the MIT License, and NQ adheres to the CC BY-SA 3.0 License. All of these licenses and agreements authorize the academic usage of their respective datasets.

A.2 Prompts Used in ActiveRAG

In this subsection, we show the prompt templates used for building our ActiveRAG model.

The prompts used in knowledge construction and cognitive nexus are shown in Table 6 and Table 7, respectively. Here {question} is the given question and {passages} indicates the retrieved passages. Besides, {chain_of_thought_reply} is filled by the output of the chain-of-thought model. Associate_knowledge_constrcution_reply}, {Anchoring_knowledge_constrcution_reply}, {Logician_knowledge_constrcution_reply}, and {Cognition_knowledge_constrcution_reply} are fed by the outcomes of the knowledge construction models of Associate, Anchoring, Logician and Cognition, respectively.

A.3 Prompts Used in Baseline Models

In Table 8, we show the prompts used in our baseline models, including vanilla ChatGPT-3.5, Guideline, Vanilla RAG, Chain-of-Note, Self-Rerank and Self-Refine. The prompt template for Chain-of-Note adheres to the prompt template described in Yu et al. (2023b), whereas the prompt template for Vanilla RAG is derived from Ram et al. (2023b).

A.4 Additional Case Studies

Additionally, we show the cases of ActiveRAG. The results of ActiveRAG using different knowledge construction methods are illustrated.

Firstly, we showcase the results in Table 9, which are generated by the ActiveRAG with the Associate model. This case illustrates that the initial chain-of-thought leads to erroneous inference. The Associate-based knowledge construction method can directly extract related knowledge to the given query. Leveraging the cognitive nexus, LLMs can discern the inadequacy of the raw chain-of-thought, identifying it as an incorrect reasoning process. By augmenting the reasoning process, LLMs yield accurate answers.

Then, we show the case of the ActiveRAG with the Anchoring model in Table 10. The ActiveRAG w. Anchoring model identifies the information regarding “Marie-Francine Hébert” as unfamiliar to the LLM. By accurately extracting the related information from passages, the LLM is able to produce the accurate response “Nantes, France”. This illustrates the functioning of Anchoring, which is designed to extract unfamiliar knowledge to assist LLMs.

Next, we present a case study of the ActiveRAG w. Logician model, as illustrated in Table 11. The Logician-based knowledge construction process generates rational knowledge from the retrieved passages. It adeptly recognizes that “The primary cause of Coretta Scott King’s death is attributed to respiratory failure resulting from complications related to ovarian cancer.” The Logician-based knowledge construction system provides accurate evidence directly, but it may be customized to address some specific questions.

Finally, we present the results of the ActiveRAG w. Cognition model in Table 12. The Cognition-based knowledge construction results illustrate that the knowledge about “ Robert Woods” does not align with the needs of the given question. This outcome underscores the significance of knowledge construction in discerning distinctions, such as differentiating the “Woods” film from the producer “Walter Wood”, thereby facilitating LLMs in generating more precise responses.

Role	Associate
Prompt	You are a cognitive scientist, to answer the following question: **Question** {question} I will provide you with several retrieved passages: **Passages:** {passages} **Task Description:** Please extract foundational knowledge that may be familiar to the model or advanced information beyond the model's already familiar foundational knowledge from these passages, and analyze the role of these contents. Summarize and consolidate these contents, which should deepen the model's understanding of the question through familiarity with these basic and advanced pieces of information. This process aims to encourage the model to comprehend the question more thoroughly and expand its knowledge boundaries.
Role	Anchoring
Prompt	You are a cognitive scientist, to answer the following question: **Question** {question} I will provide you with several retrieved passages: **Passages:** {passages} **Task Description:** Please extract content that may be unfamiliar to the model from these passages, which can provide the model with relevant background and unknown knowledge and concepts, helping it better understand the question. and analyze the role of these contents.
Role	Logician
Prompt	You are a logician, to answer the following question: **Question** {question} I will provide you with several retrieved passages: **Passages:** {passages} **Task Description:** Please extract content from these passages that can help enhance the model's causal reasoning and logical inference abilities. consolidate these contents, and analyze how the selected information may impact the improvement of the model's causal reasoning and logical inference capabilities.
Role	Cognition
Prompt	Fact-checking refers to the process of confirming the accuracy of a statement or claim through various sources or methods. This process ensures that statements or claims are based on reliable and verifiable information while eliminating inaccurate or misleading content. Fact-checking may involve examining data, literature, expert opinions, or other trustworthy sources. In the context of artificial intelligence, model illusion refers to the overconfidence response of the AI. When a model exhibits an 'illusion' (a tendency to output deceptive data), it indicates that the training data used by the model does not necessarily support the rationality of its outputs. You are a scientist researching fact-checking and addressing model illusions in artificial intelligence. To answer the following question: **Question** {question} I will provide you with several retrieved passages: **Passages:** {passages} **Task Description:** Please extract content from these passages that may be contradictory to the model's existing knowledge. Identify information that, when added, could update the model's knowledge and prevent factual errors, alleviating model illusions. Note that these passages are retrieved from the most authoritative knowledge repositories, so they are assumed to be correct.

Table 6: Prompt Templates Used in Knowledge Construction.

Role	Associate
Prompt	Here is an answer generated by a language model with the reasoning process: **Question:** {question} **Answer:** {chain_of_thought_reply} To deepen the language model's understanding of the question through familiarity with basic and advanced pieces of information. Encourage the language model to comprehend the question more thoroughly and expand its knowledge boundaries. I retrieved some foundational knowledge that is familiar to the model or advanced information beyond the language model's already familiar foundational knowledge from these passages. **knowledge:** {Associate_knowledge_constrcution_reply} Please verify the above reasoning process for errors, then enhance this reasoning process using retrieved knowledge to deepen the understanding of the question through familiarity with basic and advanced pieces of information, comprehend the question more thoroughly, and expand the knowledge boundaries. Afterward, give the answer based on the enhanced reasoning process. **Generation Format:** knowledge enhanced inference process: Answer:
Role	Anchoring
Prompt	Here is an answer generated by a language model with the reasoning process: **Question:** {question} **Answer:** {chain_of_thought_reply} To provide the language model with relevant background and unknown knowledge and concepts, helping it better understand the question. I retrieved some knowledge that is may unfamiliar with the model: **Knowledge:** {Anchoring_knowledge_constrcution_reply} Please verify the above reasoning process for errors, then enhance this reasoning process using retrieved knowledge to help it better understand the question, Afterward, give the answer based on the enhanced reasoning process. **Generation Format:** knowledge enhanced inference process: Answer:
Role	Logician
Prompt	Here is an answer generated by a language model with the reasoning process: **Question:** {question} **Answer:** {chain_of_thought_reply} To improve the language model's causal reasoning and logical inference capabilities. I retrieved some knowledge that can help enhance the language model's causal reasoning and logical inference abilities. **Knowledge:** {Logician_knowledge_constrcution_reply} Please verify the above reasoning process for errors, then enhance this reasoning process using retrieved knowledge to enhance the causal reasoning and logical inference abilities. Afterward, give the answer based on the enhanced reasoning process. **Generation Format:** knowledge enhanced inference process: Answer:
Role	Cognition
Prompt	Here is an answer generated by a language model with the reasoning process: **Question:** {question} **Answer:** {chain_of_thought_reply} To update the language model's knowledge and prevent factual errors, alleviating model illusions. I retrieved some knowledge that may update the language model's knowledge and prevent factual errors, alleviating model illusions. Note that these passages are retrieved from the most authoritative knowledge repositories, so they are assumed to be correct. I retrieved some knowledge that is may unfamiliar with the model: **Knowledge:** {Cognition_knowledge_constrcution_reply} Please verify the above reasoning process for errors, then enhance this reasoning process using retrieved knowledge to update the language model's knowledge, prevent factual errors and alleviate model illusions. Afterward, give the answer based on the enhanced reasoning process. **Generation Format:** knowledge enhanced inference process: Answer:

Table 7: Prompts Templates Used in Cognitive Nexus.

Method	Vanilla ChatGPT-3.5
Prompt	Please provide concise answers to the questions. Avoid unnecessary details: {question}
Method	Chain of Thought
Prompt	To solve the problem, Please think and reason step by step, then answer. Question: {question} Generation Format: Reasoning process: Answer:
Method	Guideline
Prompt	Prompt1: You are a knowledgeable and patient professor whose role is to guide students in solving problems correctly. Here is a question: {question} please provide a detailed analysis. Note: Since your responsibility is to guide students in answering the question, your analysis should think step by step, Please note that your role is to guide them step by step through the problem, so please don't give them the final result. Prompt2: This is another analysis of this question: {chain_of_thought_reply} Please combine this reasoning process with your reasoning processes, then verify your reasoning process and give me a better reasoning process Generation Format: inference process: Answer:
Method	Vanilla RAG
Prompt	Passages: {passages} Based on these texts, answer these questions: {question}
Method	Chain-of-Note
Prompt	Task Description: 1. Read the given question and five Wikipedia passages to gather relevant information. 2. Write reading notes summarizing the key points from these passages. 3. Discuss the relevance of the given question and Wikipedia passages. 4. If some passages are relevant to the given question, provide a brief answer based on the passages. 5. If no passage is relevant, directly provide the answer without considering the passages. Question: {question} Passage: {passages}
Method	Self-Refine
Prompt	**Prompt1:** Complete the following task: The questions that need to be answered are: {question} To answer these questions, I retrieved some passages. Passages: {passages} Task: Based on the content of the questions to be answered, Please extract relevant and useful information from these passages then provide a summary and analysis of these passages. **Prompt2:** {Same as Vanilla RAG}
Method	Self-Rerank
Prompt	**Prompt1:** Complete the following task: The questions that need to be answered are: {question} To answer these questions, I retrieved some passages. Passages: {passages} Based on the content of the questions to be answered, Please label each passage with two tags and give the reason, the first one is whether it is useful or not, and the second one is whether it is relevant or not, The four possible scenarios are as follows: 1.<useful><relevant>, 2.<useless><relevant>, 3.<useful><irrelevant>, 4.<useless><irrelevant>. Generation Format: 1. passage:..., label:..., reason:... **Prompt2:** {Same as Vanilla RAG}

Table 8: Prompts Used in Our Baseline Models.

Role: Associate
Question: Who was the first British golfer to win the US Masters? **Answer:** Sandy Lyle
Chain-of-Thought: Reasoning process: 1. The US Masters is a prestigious golf tournament held in the United States. 2. The first British golfer to win the US Masters would have had to have won it before any other British golfer. 3. The first British golfer to win the US Masters is Nick Faldo, who won in 1989 and then again in 1990. **Answer:** Nick Faldo was the first British golfer to win the US Masters.
Step 1: Passage Retrieval
Passage 1: Sandy Lyle becomes the first British player to win the U.S. Masters. Lyle sweeps his approach shot to the last green out of a fairway bunker to within, and sinks the resulting birdie putt for a one-shot victory over Mark Calcavecchia. **Passage 2:** 10 April 2013 Golfer Sandy Lyle becomes the first British winner of the US Masters. **Passage 3:** In 2007, Virginia Tech golfer Drew Weaver became the first American to win the British Amateur golf tournament since 1979. Weaver edged out 2006 Australian Amateur champion Tim Stewart and earned an invitation to the 2007 Open Championship. **Passage 4:** Gary Player of South Africa became the first foreigner to win the American Masters Tournament, taking the event by one stroke. On the very last hole, the leader, Arnold Palmer had to take six strokes.
Step 2: Knowledge Construction using Associate Agent
Foundational Knowledge: Sandy Lyle became the first British winner to win the U.S. Masters. This is a significant achievement in the history of golf, as it marked a milestone for British golfers in the prestigious tournament. Additionally, the passages mention other notable golfers and their achievements, such as Gary Player becoming the first foreigner to win the American Masters Tournament and Nicklaus winning multiple major titles in a season. **Advanced Information:** Details about other golfers' achievements, such as Drew Weaver becoming the first American to win the British Amateur golf tournament since 1979 and Ben Hogan being the only golfer to win the Masters, U.S. Open, and British Open in the same calendar year. **Analyzing the Role of Information:** By consolidating these contents, the model can deepen its understanding of the question by gaining familiarity with the historical significance of Sandy Lyle's win as the first British golfer to win the US Masters, as well as gaining knowledge about other notable achievements in golf history. This process aims to encourage the model to comprehend the question more thoroughly and expand its knowledge boundaries in the field of golf history and achievements.
Step 3: Cognitive Nexus
The reasoning process provided is incorrect. The first British golfer to win the US Masters was Sandy Lyle, not Nick Faldo. The reasoning process should be enhanced using the retrieved knowledge to deepen the understanding of the question through familiarity with basic and advanced pieces of information. Enhanced Inference Process: 1.The US Masters is a prestigious golf tournament held in the United States. 2. Sandy Lyle became the first British player to win the U.S. Masters, marking a significant milestone for British golfers in the tournament's history. 3. Other notable achievements in golf history include Gary Player becoming the first foreigner to win the American Masters Tournament and Ben Hogan being the only golfer to win the Masters, U.S. Open, and British Open in the same calendar year. **Answer:** Sandy Lyle was the first British golfer to win the US Masters. This achievement marked a significant milestone for British golfers in the history of the tournament.

Table 9: Cases Studies of ACTIVERAG w. Associate. We present one case to show the pipeline of ACTIVERAG, which includes passage retrieval, knowledge construction and cognitive nexus steps. The answers and relevant contents are highlighted.

Role: Anchoring
Question: In what city was Marie-Hélène Aubert born? **Answer:** Nantes
Chain-of-Thought: Reasoning process: 1. Marie-Hélène Aubert is a French politician and writer. 2. To find out where she was born, we can search for her birthplace in reliable sources such as her official biography or credible news articles. **Answer:** Marie-Hélène Aubert was born in Paris, France.
Step 1: Passage Retrieval
Passage 1: Marie-Francine Hébert (born March 24, 1943) is a Canadian author from Quebec. **Passage 2:** Marie Marguerite Françoise Hébert, née Goupil (1756, Paris – 1794), was a figure in the French Revolution. She was a nun in the couvent de la Conception (rue Saint-Honoré) under the name "sister of Providence", but was defrocked and on 7 February 1792 married Jacques René Hébert. **Passage 3:** Michel Pierre Alexis Hébert was born in Granville, Manche, on 7 July 1799. **Passage 4:** Marie-Hélène Aubert (born 16 November 1955 in Nantes) is a French politician and former Member of the European Parliament for the West of France. She is a member of the Socialist Party, having quit the Greens in 2008. Aubert was a Vice Chair of the ACP-EU Joint Parliamentary Assembly.
Step 2: Knowledge Construction using Anchoring Agent
Unfamiliar Knowledge: The content that may be unfamiliar to the model includes the information about Marie-Francine Hébert, a Canadian author from Quebec, and Marie Marguerite Françoise Hébert, a figure in the French Revolution who was a nun and later married Jacques René Hébert. Additionally, the mention of Michel Pierre Alexis Hébert, who was born in Granville, Manche, on 7 July 1799, may also be unfamiliar to the model. **Analyzing the role of these unfamiliar contents:** they provide background information about individuals with similar names to Marie-Hélène Aubert, but who are not the same person. This helps the model understand that the information provided in the passages is not directly relevant to the question about the birth city of Marie-Hélène Aubert. By recognizing and filtering out irrelevant information, the model can focus on the specific details related to Marie-Hélène Aubert's birth city, which is Nantes.
Step 3: Cognitive Nexus
The reasoning process provided is accurate in recognizing the unfamiliar content and its role in filtering out irrelevant information. However, the conclusion that Marie-Hélène Aubert was born in Paris, France, is incorrect based on the retrieved knowledge. Enhanced reasoning process using retrieved knowledge: 1. Marie-Hélène Aubert is a French politician and former Member of the European Parliament for the West of France. 2. The unfamiliar content about Marie-Francine Hébert, Marie Marguerite Françoise Hébert, and Michel Pierre Alexis Hébert is not directly relevant to the question about the birth city of Marie-Hélène Aubert. However, it serves to highlight the importance of distinguishing individuals with similar names to avoid confusion. **Answer:** Marie-Hélène Aubert was born in Nantes, France.

Table 10: Cases Studies of ACTIVERAG w. Anchoring. We present one case to show the pipeline of ACTIVERAG, which includes passage retrieval, knowledge construction and cognitive nexus steps. The answers and relevant contents are highlighted.

Role: Logician

Question: what did coretta scott king died from? **Answer:** Respiratory failure

Chain-of-Thought:

Reasoning process:
Coretta Scott King was the wife of civil rights leader Martin Luther King Jr. She was also an activist in her own right. She passed away in 2006 at the age of 78.
Answer: Coretta Scott King died from complications due to ovarian cancer.

Step 1: Passage Retrieval

Passage 1: Coretta Scott King began to decline in health after suffering a stroke in August 2005. She also was diagnosed with ovarian cancer...Coretta Scott King had a blood clot moving from her heart and lodge in an artery in her brain. She was hospitalized on August 16, 2005, and was set to come home as well. Alongside the physician that took care of her mother, Dr. Maggie Mermin and her sister... She became a spokesman for the American Heart Association after her mother's stroke, promoting a campaign to raise awareness about strokes.
Passage 2: Coretta Scott King died late on the evening of January 30, 2006 at the rehabilitation center in Rosarito Beach, Mexico, In the Oasis Hospital where she was undergoing holistic therapy for her stroke and advanced stage ovarian cancer. The main cause of her death is believed to be respiratory failure due to complications from ovarian cancer...
Passage 3: King's mother, Coretta Scott King, died on January 30, 2006, at the age of 78 on his 45th birthday.
Passage 4: Coretta Scott King, 78, American civil rights leader, widow of Martin Luther King, Jr., ovarian cancer.

Step 2: Knowledge Construction using Logician Agent

Consolidated Information: - Coretta Scott King died on January 30, 2006, at a rehabilitation center in Rosarito Beach, Mexico, where she was undergoing holistic therapy for her stroke and advanced-stage ovarian cancer.
- The main cause of her death is believed to be respiratory failure due to complications from ovarian cancer.
- The clinic where she died was not legally licensed to perform certain medical procedures and was operated by a controversial medical figure, Kurt Donsbach.
- King's decline in health began after suffering a stroke in August 2005 and being diagnosed with ovarian cancer.

Analyzing the Role of Information: The selected information provides a clear causal relationship between Coretta Scott King's declining health, her medical treatment, and the eventual cause of her death. By understanding the sequence of events leading to her passing, the model's causal reasoning and logical inference abilities can be improved. It can learn to recognize the impact of medical conditions, treatment choices, and the legality of medical facilities on an individual's health outcome. Additionally, the information highlights the importance of verifying the credentials and licensing of medical facilities and practitioners, contributing to the model's ability to make informed judgments about healthcare practices and their potential consequences.

Step 3: Cognitive Nexus

Enhanced reasoning process:
1. Coretta Scott King passed away on January 30, 2006, at the rehabilitation center in Rosarito Beach, Mexico, where she was undergoing holistic therapy for her stroke and advanced-stage ovarian cancer. 2. Her declining health began after suffering a stroke in August 2005 and being diagnosed with ovarian cancer.
3. The main cause of her death is believed to be respiratory failure due to complications from ovarian cancer, which led to her hospitalization and ongoing care.
4. Additionally, she had a blood clot move from her heart and lodge in an artery in her brain, which also contributed to her declining health and ongoing medical issues.
5. The clinic at which she died, Hospital Santa Monica (licensed as Clinica Santo Tomas), was shut down due to operating without proper licensing and controversial medical practices.
Answer: Coretta Scott King died from respiratory failure due to complications from advanced-stage ovarian cancer, exacerbated by the impact of a stroke.

Table 11: Cases Studies of ACTIVERAG w. Logician. We present one case to show the pipeline of ACTIVERAG, which includes passage retrieval, knowledge construction and cognitive nexus steps. The answers and relevant contents are highlighted.

Role: Cognition
Question: Who was the producer of The Woods? **Answer:** Matthew Lessner
Chain-of-Thought: 1. The Woods is a television series. 2. To find the producer, we can look at the credits of the show or search online for information about the production team. **Answer:** The producer of The Woods is Bartosz Kowalski.
Step 1: Passage Retrieval
Passage 1: Walter Wood (producer). **Passage 2:** The Woods is a 2011 film written and directed by Matthew Lessner and starring Toby David and Justin Phillips. The script was written by Lessner with contributing writer Adam Mortemore, and additional dialogue by David and Phillips. **Passage 3:** "Robert Woods is a classical music producer. Woods is a founder and former president of the independent record label Telarc International Corporation. **Passage 4:** "The Wood is a 1999 American coming of age film directed by Rick Famuyiwa and starring Omar Epps, Richard T. Jones and Taye Diggs. It was written by Rick Famuyiwa and Todd Boyd.
Step 2: Knowledge Construction using Cognition Agent
Contradictory Knowledge: The contradictory information in the passages is the mention of Robert Woods as a classical music producer and founder of Telarc International Corporation, which does not align with the question about the producer of the film "The Woods". **Analyzing the Role of Information:** This information could update the model's knowledge and prevent factual errors by adding the correct producer of "The Woods" film, which is Matthew Lessner."
Step 3: Cognitive Nexus
knowledge enhanced inference process: 1. The contradictory information in the passages is the mention of "Robert Woods" as a classical music producer, which does not align with the question about the producer of the film "The Woods." 2. The correct producer of "The Woods" film is Matthew Lessner, as mentioned in the passage "The Woods is a 2011 film written and directed by Matthew Lessner. 3. This information could update the model's knowledge and prevent factual errors by adding the correct producer of "The Woods" film, which is Matthew Lessner. **Answer:** The producer of The Woods is Matthew Lessner.

Table 12: Example of ACTIVERAG w. Cognition. We present one case to show the pipeline of ACTIVERAG, which includes passage retrieval, knowledge construction and cognitive nexus steps. The answers and relevant contents are highlighted.